

Technical Requirements - Intelligent Invoice Exception Resolution

Current Architecture & Systems

System Architecture: Hybrid monolithic and service-oriented - **Core ERP:** SAP S/4HANA (on-premise) - handles invoice processing, PO management, GRN recording - **Email System:** Microsoft Exchange Server 2019 (on-premise) with Office 365 hybrid - **Document Storage:** Network-attached storage (NAS) for invoice PDFs and supporting documents - **Reporting:** SAP BusinessObjects for financial reporting

Technology Stack: - **Application Layer:** Java-based custom extensions on SAP - **Database:** SAP HANA (in-memory database) - **Integration Middleware:** SAP PI/PO (Process Integration/Orchestration) - **Frontend:** SAP Fiori for modern UI, legacy SAP GUI still in use

Cloud Maturity: Early hybrid stage - Office 365 for email and collaboration (cloud) - Core ERP remains on-premise due to historical concerns about data sovereignty - No experience with AWS or other cloud platforms - IT leadership now open to cloud-native solutions for new capabilities

Integration Landscape

Systems Requiring Integration:

1. **SAP S/4HANA** (Critical)
 - Invoice data, PO data, GRN data
 - Vendor master data
 - Approval workflows
 - Integration method: SAP OData APIs (REST-like)
2. **Microsoft Exchange/Office 365** (Critical)
 - Read incoming supplier emails
 - Send outgoing emails to suppliers and internal staff
 - Search email history for context
 - Integration method: Microsoft Graph API (REST)
3. **SharePoint** (Important)
 - Contract documents and supplier agreements
 - Historical correspondence archives
 - Integration method: Microsoft Graph API
4. **Procurement System** (Important)
 - Supplier performance data
 - Contract terms and conditions
 - Integration method: Custom REST API
5. **Document Management** (Important)
 - Invoice PDFs and supporting documents
 - Integration method: File system access or S3 migration

Integration Patterns: - **Current:** Primarily batch-based integration via SAP PI/PO (nightly jobs) - **Required for Agents:** Real-time event-driven integration - Invoice discrepancy detected → trigger agent investigation - Supplier email received → trigger agent response - GRN created → check for pending invoice exceptions

Performance Requirements: - **Invoice volume:** 50,000/month (2,500/day average, 5,000/day peak at month-end) - **Exception volume:** 15,000/month (750/day average) - **Response time:** Agent should begin investigation within 5 minutes of exception detection - **Email response:** Agent should respond to supplier emails within 2 hours during business hours - **Availability:** 99.5% during business hours (7am-7pm AEST, Monday-Friday)

Critical SLAs: - Month-end close: All exceptions must be resolved within 3 business days - High-value invoices (>\$50K): 24-hour resolution target - Supplier payment terms: Must not breach payment deadlines

Data Strategy & Readiness

Data Sources:

1. **Structured Data (SAP HANA):**
 - Invoice records: invoice number, amount, date, vendor ID, PO reference
 - Purchase orders: PO number, line items, quantities, prices
 - GRNs: receipt date, quantities received, quality status
 - Vendor master: vendor ID, name, contact details, payment terms, performance ratings
2. **Unstructured Data:**
 - **Email correspondence:** 2 years of email history (~500K emails)
 - **Invoice PDFs:** 7 years of scanned invoices (compliance requirement)
 - **Contract documents:** PDF contracts with supplier terms
 - **Spreadsheet tracking:** Excel files with tribal knowledge and exception notes

Data Quality: - **Structured data:** High quality (95%+ accuracy) - SAP enforces data validation - **Unstructured data:** Variable quality - Email threads often fragmented across multiple conversations - Inconsistent naming conventions for attachments - Tribal knowledge in spreadsheets not standardized

Data Sensitivity: - **Confidential:** Vendor bank account details, pricing agreements, contract terms - **Internal:** Invoice amounts, PO numbers, approval decisions - **PII:** Vendor contact names, email addresses, phone numbers

Data Residency: All data must remain within Australia (regulatory requirement)

Knowledge Base Requirements: - **Supplier-specific rules:** 200+ undocumented rules (e.g., “Supplier X always ships 2% over”) - **Approval thresholds:**

5 different workflows based on amount and risk - **Historical patterns:** Need to learn from 2 years of exception resolution history - **Contract terms:** Extract acceptable variance clauses from PDF contracts

Vector Database Needs: - Semantic search across email history to find similar past exceptions - Contract clause retrieval for variance acceptance rules - Supplier communication pattern analysis

Model & AI Infrastructure

Current AI/ML Experience: Limited - No production AI/ML models currently deployed - Previous chatbot pilot used Azure Bot Framework (failed due to poor accuracy) - IT team has basic understanding of AI concepts but no hands-on experience - No MLOps practices or infrastructure

Foundation Model Requirements:

1. **Document Understanding:** Extract data from invoice PDFs and contracts
2. **Email Comprehension:** Understand supplier email intent and context
3. **Reasoning & Decision-Making:** Evaluate whether variance is acceptable based on multiple factors
4. **Email Generation:** Draft professional, contextually appropriate emails
5. **Pattern Recognition:** Learn from historical exception resolutions

Model Preferences: - Prefer AWS-native solutions (Bedrock) to minimize vendor complexity - Need explainable AI - must be able to explain why agent made a decision - Require guardrails to prevent inappropriate decisions or communications

Prompt Engineering: - No in-house expertise currently - Need templates and best practices for agent prompts - Require ability to refine prompts based on agent performance

Model Governance Needs: - Version control for agent configurations and prompts - A/B testing capability to compare agent performance - Rollback capability if agent performance degrades

Agent Architecture Requirements

Required Agent Roles:

1. **Investigator Agent**
 - Analyzes invoice vs GRN discrepancies
 - Searches email history for context
 - Retrieves contract terms and supplier history
 - Determines if variance is within acceptable bounds
2. **Communication Agent**
 - Drafts emails to suppliers requesting clarification

- Drafts internal emails to procurement team
 - Adjusts tone based on supplier relationship (formal vs casual)
 - Handles follow-up emails if no response received
3. **Decision Agent**
 - Evaluates whether to accept variance, reject invoice, or escalate
 - Considers multiple factors: contract terms, supplier history, item criticality, amount
 - Provides explanation for decision
 - Identifies when human judgment is required
 4. **Escalation Agent**
 - Determines escalation criteria (high-value, high-risk, policy violation)
 - Routes to appropriate approver (AP manager, procurement, finance director)
 - Provides full context and recommendation to human reviewer
 5. **Learning Agent**
 - Captures patterns from resolved exceptions
 - Updates knowledge base with new supplier-specific rules
 - Identifies process improvement opportunities
 - Monitors agent performance and suggests refinements

Agent Coordination: - Investigator Agent triggers Communication Agent when clarification needed - Communication Agent waits for supplier response, then triggers Decision Agent - Decision Agent triggers Escalation Agent when criteria met - All agents log to Learning Agent for continuous improvement

Autonomy Boundaries: - **Fully autonomous:** Variances <5% for trusted suppliers, standard items - **Autonomous with notification:** Variances 5-10%, notify AP manager after decision - **Human-in-the-loop:** Variances >10%, high-value items (>\$50K), new suppliers, policy violations

Security & Identity

Authentication & Authorization: - **Current:** Active Directory for user authentication - **Required:** - Service accounts for agent access to SAP and Exchange - OAuth 2.0 for Microsoft Graph API integration - API keys for SAP OData services - Role-based access control for agent capabilities

Secrets Management: - Currently stored in configuration files (not secure) - Need centralized secrets management (AWS Secrets Manager preferred)

Encryption: - **At rest:** All data must be encrypted (AES-256) - **In transit:** TLS 1.2 minimum for all API calls - **In use:** Sensitive data (bank accounts, pricing) should be masked in logs

Network Security: - **Current:** On-premise systems behind corporate firewall - **Required:** - VPN or AWS PrivateLink for secure cloud-to-on-premise connectivity - Network segmentation for agent infrastructure - Egress controls to prevent data exfiltration

Audit Logging: - All agent actions must be logged with timestamps and user context - Logs must be immutable and retained for 7 years (compliance) - Need to log: decision rationale, data accessed, emails sent, escalations

Observability & Operations

Current Monitoring: - Basic SAP monitoring (system health, database performance) - Email server monitoring (uptime, queue length) - No application performance monitoring (APM) - Manual exception tracking in spreadsheets

Required Observability:

1. **Agent Performance Metrics:**
 - Resolution time per exception
 - Autonomous resolution rate
 - Decision accuracy (validated by human review)
 - Escalation rate and reasons
 - Supplier response rate to agent emails
2. **System Health Metrics:**
 - Agent availability and response time
 - API call latency (SAP, Exchange)
 - Model inference time
 - Queue depth for pending exceptions
3. **Business Metrics:**
 - Exceptions resolved per day
 - Cost per exception (token usage)
 - AP team time saved
 - Vendor satisfaction (from surveys)

Alerting Requirements: - Agent failure or degraded performance - High escalation rate (>30% indicates agent struggling) - SLA breach risk (exceptions approaching deadline) - Security incidents (unauthorized access, data anomalies)

Operational Runbooks: - Need documented procedures for common issues - Escalation path for agent failures - Rollback procedures if agent performance degrades

Scalability & Performance

Current Volumes: - 50,000 invoices/month - 15,000 exceptions/month (30% exception rate) - 750 exceptions/day average, 1,500/day at month-end peak

Growth Projections: - 20% annual growth in invoice volume - Expect exception rate to decrease to 20% as agents improve processes

Performance Targets: - **Investigation time:** <5 minutes per exception - **Email response time:** <2 hours during business hours - **End-to-end resolution:** <8 hours for 70% of exceptions - **Concurrent processing:** Handle 100 exceptions simultaneously at peak

Auto-scaling Requirements: - Scale up during month-end peaks (2x capacity)
- Scale down during low-volume periods to control costs - No experience with auto-scaling, need guidance

Disaster Recovery: - **RTO:** 4 hours (can tolerate brief outage) - **RPO:** 1 hour (acceptable to re-process recent exceptions) - Need backup of agent knowledge base and configuration

Development & Deployment

Current Practices: - **Version Control:** Git (Bitbucket) for custom SAP code - **CI/CD:** Basic Jenkins pipelines for SAP transports - **Testing:** Primarily manual UAT, limited automated testing - **Environments:** Dev, QA, Production (no staging environment) - **Deployment Frequency:** Monthly releases for SAP changes

Required for Agentic AI:

1. **Infrastructure as Code:**
 - No current IaC experience
 - Prefer AWS CDK or CloudFormation
 - Need templates and best practices
2. **Agent Deployment:**
 - Containerized agents (Docker/ECS preferred)
 - Blue-green deployment for zero-downtime updates
 - Ability to roll back agent versions quickly
3. **Testing Strategy:**
 - Unit tests for agent tools and integrations
 - Integration tests with SAP and Exchange (test environments)
 - **Agent evaluation:** Test agent decisions against historical data (2 years of exceptions)
 - A/B testing to compare agent versions
4. **Environment Strategy:**
 - **Dev:** For agent development and prompt engineering
 - **Test:** Integration testing with SAP/Exchange test systems
 - **Staging:** Full production-like environment for UAT
 - **Production:** Live agent processing
5. **Monitoring & Feedback Loop:**
 - Real-time agent performance dashboard
 - Weekly agent performance reviews
 - Monthly agent capability updates based on learnings

Integration Architecture

Event-Driven Architecture: - **Trigger Events:** - Invoice posted in SAP with discrepancy flag → Trigger Investigator Agent - Supplier email received → Trigger Communication Agent - GRN created → Check for pending invoice

exceptions - Escalation approved/rejected → Update agent knowledge base

Message Queue Requirements: - Need reliable message queue for agent coordination (SQS preferred) - Dead letter queue for failed agent tasks - Priority queue for high-value exceptions

API Gateway: - Centralized API gateway for agent-to-system communication - Rate limiting to prevent overwhelming SAP/Exchange - Request/response logging for audit trail

Data Synchronization: - Real-time sync of invoice, PO, GRN data from SAP - Email sync from Exchange (near real-time, 5-minute polling acceptable) - Vendor master data sync (daily batch acceptable)

Additional Technical Details

SAP API Capabilities: - **OData Services Available:** - Invoice API: Read invoice headers, line items, status - PO API: Read purchase orders, line items, approval status - GRN API: Read goods receipts, quantities, dates - Vendor API: Read vendor master data, payment terms, contact info - **Authentication:** OAuth 2.0 with service account credentials - **Rate Limits:** 1,000 API calls per hour per service account (sufficient for our volume) - **Response Time:** Average 200ms per API call

Exchange/Graph API Permissions: - **Required Permissions:** - Mail.Read: Read emails from shared AP mailbox - Mail.Send: Send emails on behalf of AP team - Mail.ReadWrite: Mark emails as read, move to folders - **Authentication:** OAuth 2.0 with application permissions (not delegated) - **Rate Limits:** 10,000 API calls per 10 minutes (sufficient for our volume) - **Mailbox:** Shared mailbox (ap@company.com) for all supplier communication

Network Topology: - **On-Premise:** - SAP S/4HANA: 10.0.1.0/24 subnet - Exchange Server: 10.0.2.0/24 subnet - Corporate firewall: Allows outbound HTTPS (443) only - **AWS:** - VPC: 172.16.0.0/16 - Private subnets: 172.16.1.0/24, 172.16.2.0/24 (agents, Lambda) - Public subnets: 172.16.101.0/24, 172.16.102.0/24 (NAT gateways) - **Connectivity:** - AWS Site-to-Site VPN for SAP connectivity - Direct internet for Exchange/Office 365 (already cloud-based) - No PrivateLink required (Office 365 is public endpoint)

Existing AWS Account: - Have AWS account (created 6 months ago for exploration) - Currently only using S3 for backups (minimal usage) - No production workloads on AWS yet - **Approved Services** (from IT governance): - Compute: Lambda, ECS (Fargate preferred) - Storage: S3, EFS - Database: DynamoDB, RDS (Aurora) - AI/ML: Bedrock, SageMaker - Networking: VPC, VPN, CloudFront - Security: IAM, Secrets Manager, KMS - Monitoring: CloudWatch, X-Ray - **Restricted Services:** EC2 (prefer serverless), Redshift (not needed)

Data Migration Plan: - **Email History** (2 years, ~500K emails): - Export

from Exchange to PST files (1 week) - Convert PST to EML format (automated tool, 2 days) - Upload to S3 (1 day) - Index in OpenSearch for semantic search (3 days) - **Total time:** 2 weeks - **Cost:** \$500 (storage) + \$1,000 (indexing compute) - **Exception History** (2 years, ~360K exceptions): - Export from SAP to CSV (1 day) - Transform and load to DynamoDB (2 days) - **Total time:** 3 days - **Cost:** \$200 (storage) - **Invoice PDFs** (7 years, ~4.2M invoices): - Already in NAS, migrate to S3 over 4 weeks (bandwidth limited) - Use S3 Transfer Acceleration for faster upload - **Cost:** \$10,000 (storage) + \$2,000 (transfer)

Token Cost Modeling (Detailed): - **Per Exception Investigation:** - Read invoice/PO/GRN data: 500 tokens (structured data) - Search email history: 2,000 tokens (semantic search + context) - Retrieve contract terms: 1,500 tokens (PDF extraction) - Decision reasoning: 2,500 tokens (multi-step reasoning) - Email generation: 1,000 tokens (draft + refinement) - **Total per exception:** 7,500 tokens average - **Monthly Volume:** - 15,000 exceptions $\times$ 7,500 tokens = 112.5M tokens - Bedrock Nova Pro pricing: \$0.008 per 1K input tokens, \$0.032 per 1K output tokens - Assume 70% input, 30% output: $(112.5M \times 0.7 \times \$0.008) + (112.5M \times 0.3 \times \$0.032) = \$630 + \$1,080 = \$1,710/\text{month}$ - **Annual token cost:** \$20,520 - **Knowledge Base Retrieval:** - 15,000 queries $\times$ \$0.10 per query = \$1,500/month - **Annual cost:** \$18,000 - **Total AI/ML Cost:** \$38,520/year (vs \$60,000 budgeted, good buffer)

Backup SAP Access: - If agents fail, AP team can continue manual processing - SAP GUI and Fiori interfaces remain available - Email access via Outlook (not affected by agent failure) - Spreadsheet tracking can be resumed if needed - **Downtime tolerance:** 4 hours (can catch up on backlog)

Email Sending Limits: - **Exchange Online:** 10,000 emails per day per mailbox - **Current volume:** 15,000 exceptions/month = 500/day average, 1,000/day peak - **Agent email volume:** ~30% of exceptions require email (150/day average, 300/day peak) - **Conclusion:** Well within limits, no throttling concerns

Contract Parsing Complexity: - **Contract formats:** 80% PDF (scanned), 20% Word (digital) - **OCR quality:** Variable (70-95% accuracy depending on scan quality) - **Structure:** Semi-structured (standard clauses but varied formatting) - **Key clauses to extract:** - Acceptable variance thresholds (e.g., “ $\pm 5\%$ quantity variance allowed”) - Payment terms (e.g., “Net 30 days, 2% discount if paid within 10 days”) - Delivery terms (e.g., “FOB, partial deliveries allowed”) - **Approach:** Use Bedrock Data Automation with custom blueprint for contract extraction - **Fallback:** If confidence $< 70\%$, escalate to human for manual review

Supplier Notification Requirements: - **Legal review:** Required before sending AI disclosure emails (2 weeks, \$5K legal fees) - **Privacy laws:** Australian Privacy Act requires disclosure of automated decision-making - **Recommendation:** Transparent disclosure to all suppliers - **Opt-out:** Suppliers can request human-only communication (estimated $< 5\%$, ~10 suppliers) - **Implementation:**

Flag in vendor master data, agents skip these suppliers